



East West University
Department of Computer Science and Engineering
Course Outline
Spring 2026 Semester

Course: CSE209/ICE109 Electrical Circuits (Section 7 & 8)

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	2 Hours/Week for 13 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

Prerequisite

None

Instructor Information

Instructor: **Musharrat Khan**
Senior Lecturer, Department of Computer Science and Engineering
Office: Room # 636
Tel. No.: 09666775577 (hunting) ext. 329
E-mail: musharrat.khan@ewubd.edu
TA: TBA

Class Routine & Office Hours

Day	10:10-11:40	11:50-01:20	01:30-03:00	03:10-04:40
Sunday	CSE209(8) Room: AB2-601	Office Hour	CSE345(5) Room: AB2-701	
Monday	CSE209(7) Room: AB2-601	Office Hour	CSE345(5) LAB Room: 634	
Tuesday	CSE209(8) Room: AB2-601	Office Hour	CSE345(5) Room: AB2-701	
Wednesday	CSE209(7) Room: AB2-601	Office Hours	CSE251(8) LAB Room: 547	
Thursday	CSE209(8) LAB Room: 547	Office Hour	CSE209(7) LAB Room: 449	Office Hour

Course Objective

This course provides the students with fundamental knowledge of electrical circuits and various techniques of analyzing them. This course also provides hands-on experience in building and testing electrical circuits. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE251 Electronic Circuits, CSE345 Digital Logic Design, CSE350 Data Communications, CSE360 Computer Architecture, CSE442 Microprocessor and Microcontroller, and CSE490 VLSI design.

Knowledge Profile

K1: Theory-based natural sciences

K3: Theory-based engineering fundamentals

Learning Domains

Cognitive - C2: Understanding, C3: Applying

Psychomotor - P2: Manipulation, P3: Precision

Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required

EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	EP/EA
CO1	Discuss and use the concepts of electrical circuit elements, circuit variables, circuit laws, and circuit combinations for solving DC and AC circuits.	PO1	C2, C3	K1	-
CO2	Examine different circuit analysis techniques for DC circuit solution.	PO2	C3	K3	-
CO3	Examine different circuit analysis techniques for AC circuit solution.	PO2	C3	K3	-
CO4	Use analytical, software, and hardware tools and techniques; perform and demonstrate skills; and write report for designing, building, and testing electrical circuits.	PO1	C3 P2, P3 A2	K1, K3	EP1, EP2

Lecture Plan, Course Topics, Teaching-Learning Method, and Assessment Scheme

Lec. Plan (L)	Subtopics/Details	Course Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels		CO Mark	Exam Mark
					C2	C3		
L1	Introduction to Electrical Circuits; System of units; Charge, Current, Voltage; Circuit elements: Active & Passive elements, Independent and dependent types of Dependent sources	Basic Concepts of DC circuit, charge and current, voltage, power and energy, Circuit elements. Ohm's Law, Kirchhoff's Voltage Law, Kirchhoff's Current Law, Series-Parallel connections, Voltage and Current Division, Wye-delta Transformations.	Lecture, Class Discussion, Discussion outside class with Instructor/TA	CO1	5	5	10	Mid (30)
L2	Power and energy in DC circuits; Passive sign convention; Problem solving							
L3	Ohm's Law; Resistance and Conductance; Formulas to calculate power across resistor; Concepts of open circuit and short circuit; Nodes, Branches, Loops and Mesh in a circuit							
L4	Kirchhoff's Voltage Law (KVL); Kirchhoff's Current Law (KCL); Problem solving							

L5	Series and Parallel resistor connections; Voltage and Current Division; Problem solving						
L6	Equivalent resistance; Wye–Delta transformations; Problem solving						
L7	Introduction to DC circuit analysis; Nodal analysis with current sources; Problem solving	DC Circuit Analysis: Nodal method (including independent and dependent sources, Super-node analysis).	Do	CO2		6	6
L8	Nodal analysis with voltage sources; Super-node analysis; Problem solving						
L9	Mesh analysis with voltage sources; Mesh analysis with current sources; Problem solving	DC Circuit Analysis: Mesh method (including independent and dependent sources, Super-mesh analysis).	Do	CO2		6	6
L10	Super-mesh analysis; Problem solving; Nodal vs. Mesh analysis						
L11	Introduction to circuit theorems; Linearity property; Superposition theorem; Applications; Problem solving	DC Circuit Analysis: Superposition, Source Transformation	Do	CO2		8	8

L12	Class Test 1 CO1: 5 (L1-L6) CO2: 5 (L7-L10) Total: 10 Marks Source transformation; Problem solving							
L13	Mid Term Examination (L1–L12)							
L14	Thevenin's theorem; Procedure and applications; Problem solving	DC Circuit Analysis: Thevenin and Norton's equivalents, Maximum power transfer	Do	CO2		7	7	Final (30)
L15	Norton's theorem; Procedure and applications; Problem solving							
L16	Maximum power transfer theorem; Problem solving							
L17	Introduction to AC circuits; Sinusoids; Phasors; Phasor relationships for Circuit Elements; Impedance and admittance	Basic Concepts of AC circuit, Sinusoids, Phasors, Phasor relationships for Circuit Elements, Impedance and admittance. Kirchhoff's laws in Frequency domain, Impedance Combinations.	Do	CO1	5		5	
L18	Kirchhoff's laws in frequency domain; Impedance combinations							
L19	AC circuit analysis using basic laws							
L20	AC circuit analysis using advanced analysis techniques: Nodal and Mesh		Do	CO3		18	18	

L21	AC circuit analysis using circuit theorems: Superposition and Source Transformation Thevenin and Norton equivalents	AC Circuit Analysis: Nodal and Mesh Analysis, Superposition, Source Transformation, Thevenin and Norton equivalents. Instantaneous and average power; Maximum average power transfer, Effective or RMS value, apparent power and power factor, Complex power.						
L22	Instantaneous and average power; Maximum average power transfer; Problem solving							
L23	Effective or RMS value; Problem solving Apparent power and power factor;							
L24	Class Test 2 CO3: 5 (L20-L22) Complex power; Problem solving							
L25	Review and Discussion							
L26	Final Examination (L14–L24)							

Laboratory Experiments and Assessment Scheme

Experiment	Teaching-Learning Method	CO	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
			C3	P2	P3	A2	
Introduction to Circuit Elements and Variables	Preparing Pre-Lab Report, Lab Experiment, and Result Analysis	CO4					

Series DC Circuit and Verification of Kirchhoff's Voltage Laws	Do	CO4					
Parallel DC Circuit and Verification of Kirchhoff's Current Laws	Do	CO4					
Bias Point Detail Analysis of DC Circuit with Independent Sources Using PSpice Schematics.	Do	CO4					
Bias Point Detail Analysis of DC Circuit with Dependent Sources Using PSpice Schematics.	Do	CO4					
Verification of Superposition Theorem	Do	CO4					
Verification of Thevenin's theorem	Do	CO4					
DC Circuit Analysis in PSpice using Source and Resistance Sweep	Do	CO4					
Experimental Study of Sinusoids and Their Characteristics	Do	CO4					
AC Circuit Analysis using	Do	CO4					

PSpice Schematics							
Lab Performance		CO4	4	1	0	0	5
Lab Exam		CO4	7	1	1	1	10
Total			11	2	1	1	15

Mini Project

Mini Project	Teaching-Learning Method	CO	EP/EA	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
				C3	P2	P3	A2	
Lab-based Mini Project including Report and Presentation	Group-based, moderately complex electrical circuit building for practical application with report writing and presentation	CO4	EP1, EP2	2	1	1	1	5

Overall Assessment Scheme

Assessment Area	CO				Other	Total	PO Marks	
	CO1	CO2	CO3	CO4			PO1	PO2
Class Participation					5	5		
Class Test/Quiz	5	5	5			15	5	10
Mid Semester Exam	10	20				30	10	20
Final Exam	5	7	18			30	5	25
Laboratory Performance and Lab Exam				15		15	15	
Mini Project				5		5	5	
Total	20	32	23	20		100	40	55

Teaching Materials/Equipment

Text book:

1. Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits*, 5th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2008.

Teaching-Learning Method:

Lectures, Discussions, Lab Exercises, and post-lab assignments

Lab Manual:

Lab manual will be provided.

Mini Project Description:

Mini Project description will be provided.

Equipment/Software:

Circuit trainer board, Power supply, Circuit elements, Voltmeter, Ammeter, Multimeter and PSpice Software.

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% - 79%	A	3.75
70% - 74%	A-	3.50
65% - 69%	B+	3.25
60% - 64%	B	3.00
55% - 59%	B-	2.75
50% - 54%	C+	2.50
45% - 49%	C	2.25
40% - 44%	D	2.00
Less than 40%	F	0.00

Exam Dates

Section	Class Slot	Mid Semester Exam	Final
7	MW	04.03.2026	29.04.2026
8	ST	01.03.2026	26.04.2026

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative

exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss any exam, the student **MUST** get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application.
- For **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones **MUST** be turned to silent mode during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**